

Section 5.1 Exercises

Directions: For exercises 1 - 16, draw an angle in standard position with the given measure.

1. 30°

2. 300°

3. -80°

4. 135°

5. -150°

6. $\frac{2\pi}{3}$

7. $\frac{7\pi}{4}$

8. $\frac{5\pi}{6}$

9. $\frac{\pi}{2}$

10. $-\frac{\pi}{10}$

11. 415°

12. -120°

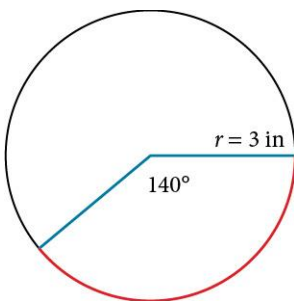
13. -315°

14. $\frac{22\pi}{3}$

15. $-\frac{\pi}{6}$

16. $-\frac{4\pi}{3}$

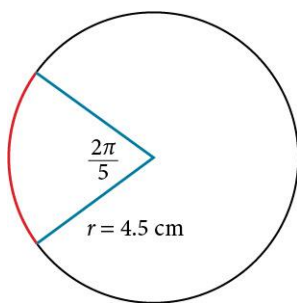
Directions: Use the picture below for exercises 17 - 18. Round your answer to two decimal places.



17. Find the arc length.

18. Find the area of the sector.

Directions: Use the picture below for exercises 19 - 20. Round your answer to two decimal places.



19. Find the arc length.

20. Find the area of the sector.

Directions: For exercises 21 - 27, convert the angles from radians to degrees.

21. $\frac{3\pi}{4}$

22. $\frac{\pi}{9}$

23. $-\frac{5\pi}{4}$

24. $\frac{\pi}{3}$

25. $-\frac{7\pi}{3}$

26. $-\frac{5\pi}{12}$

27. $\frac{11\pi}{6}$

Directions: For exercises 28 - 34, convert the angles from degrees to radians.

28. 90°

29. 100°

30. -540°

31. -120°

32. 180°

33. -315°

34. 150°

Directions: For exercises 35 - 40, use the given information to find the length of a circular arc. Round your answer to two decimal places.

35. Find the length of the arc of a circle of radius 12 inches subtended by a central angle of $\frac{\pi}{4}$ radians.
36. Find the length of the arc of a circle of radius 5.02 miles subtended by the central angle of $\frac{\pi}{3}$.
37. Find the length of the arc of a circle of diameter 14 meters subtended by the central angle of $\frac{5\pi}{6}$.
38. Find the length of the arc of a circle of radius 10 centimeters subtended by the central angle of 50° .
39. Find the length of the arc of a circle of radius 5 inches subtended by the central angle of 220° .
40. Find the length of the arc of a circle of diameter 12 meters subtended by the central angle is 63° .

Directions: For exercises 41 - , use the given information to find the area of the sector. Round your answer to two decimal places.

41. A sector of a circle has a central angle of 45° and a radius 6 cm.
42. A sector of a circle has a central angle of 30° and a radius of 20 cm.
43. A sector of a circle with diameter 10 feet and an angle of $\frac{\pi}{2}$ radians.
44. A sector of a circle with radius of 0.7 inches and an angle of π radians.

Directions: For exercises 45 - 48, find the angle between 0° and 360° that is coterminal to the given angle.

45. -40°
46. -110°
47. 700°
48. 1400°

Directions: For exercises 49 - 52, find the angle between 0 and 2π in radians that is coterminal to the given angle.

49. $-\frac{\pi}{9}$
50. $\frac{10\pi}{3}$
51. $\frac{13\pi}{6}$
52. $\frac{44\pi}{9}$